

AI Meeting Notes Analyzer using LangGraph

Objective

Build an AI-powered Meeting Notes Analyzer that converts a meeting transcript into structured insights using a LangGraph workflow. The system should analyze the transcript and automatically generate:

- Key discussion topics
- Summary of the meeting
- Action items
- Task owners
- Priority level of tasks

Github link –

<https://github.com/Priyanka-chila/meeting-analyzer-agent>



Project Objective

In real-world organizations, meetings generate a large amount of information, but important decisions, action items, and ownership can easily be missed.

This project addresses that problem by automatically analyzing meeting transcripts and generating structured meeting notes.

Input

A meeting transcript such as:

Product Manager: We have received several complaints about the mobile app crashing during login.

Mobile Developer: I will review the Android login module.

QA Tester: I will prepare a detailed bug report.

Backend Developer: I will verify the authentication API.

Output

Meeting Summary

The team discussed a critical Android login crash affecting multiple customers and investigated potential authentication API and mobile application issues.

Key Topics

1. Android login crash
2. Authentication API
3. Backend deployment
4. Root cause investigation

Action Items

1. Review Android login module

Owner: Mobile Developer

Priority: High

2. Prepare detailed bug report

Owner: QA Tester

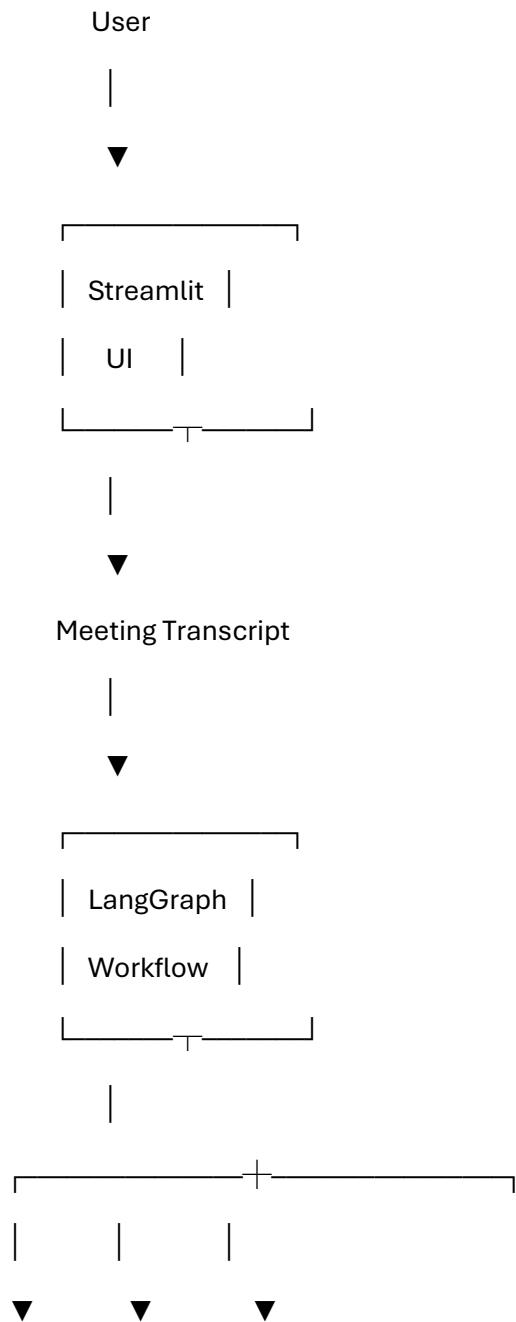
Priority: High

3. Verify authentication API

Owner: Backend Developer

Priority: High

 Architecture



Topic Agent Summary Agent Action Agent

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Priority Agent

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Final Report Agent

LangGraph Workflow

The application uses a multi-step LangGraph workflow.

START

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Input Transcript

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Topic Extraction Agent

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Meeting Summary Agent

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Action Item Extraction Agent

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Action Items Found?



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▼ Yes

Priority Classification Agent

|

▼

Final Output

|

▼

END

Agents

1. Topic Extraction Agent

Identifies the main discussion topics from the transcript.

2. Meeting Summary Agent

Creates a concise 3–5 sentence summary of the meeting.

3. Action Item Agent

Extracts:

- Task description
- Task owner

If an owner is not mentioned:

Owner: Not specified

4. Priority Agent

Classifies task priority based on urgency and deadlines.

Supported levels:

High

Medium

Low

The agent considers phrases such as:

- urgently
- immediately
- today
- by tomorrow
- this week
- within the next hour
- critical

5. Conditional Routing

If no action items are detected, the priority agent is skipped.

Action Items

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|— None —► Skip Priority Agent

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|— Found —► Priority Agent